

$$a) \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$$

pravá strana:

$$\begin{aligned} \frac{1}{4} (\langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle - \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle) & \stackrel{*}{=} \frac{1}{4} (\langle \vec{u}, \vec{u} + \vec{v} \rangle + \langle \vec{v}, \vec{u} + \vec{v} \rangle - (\langle \vec{u}, \vec{u} - \vec{v} \rangle - \langle \vec{v}, \vec{u} - \vec{v} \rangle)) = \\ \underbrace{\|\vec{u} + \vec{v}\|^2}_{\langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle} & \stackrel{**}{=} \frac{1}{4} (\cancel{\langle \vec{u}, \vec{u} \rangle} + \langle \vec{u}, \vec{v} \rangle + \cancel{\langle \vec{v}, \vec{u} \rangle} + \langle \vec{v}, \vec{v} \rangle - (\cancel{\langle \vec{u}, \vec{u} \rangle} - \langle \vec{u}, \vec{v} \rangle - \langle \vec{v}, \vec{u} \rangle + \cancel{\langle \vec{v}, \vec{v} \rangle})) = \\ & = \frac{1}{4} (2\langle \vec{u}, \vec{v} \rangle + 2\langle \vec{v}, \vec{u} \rangle) = \frac{1}{4} (2\langle \vec{v}, \vec{v} \rangle + 2\overline{\langle \vec{u}, \vec{v} \rangle}) \stackrel{***}{=} \frac{1}{4} \cdot 4 \operatorname{Re} \langle \vec{u}, \vec{v} \rangle = \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) \end{aligned}$$

$$b) \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2)$$

pravá strana:

$$\begin{aligned} \frac{1}{4} (\langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle - \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle) & \stackrel{*}{=} \frac{1}{4} (\langle \vec{u}, \vec{u} - i\vec{v} \rangle - \langle i\vec{v}, \vec{u} - i\vec{v} \rangle - (\langle \vec{u}, \vec{u} + i\vec{v} \rangle + \langle i\vec{v}, \vec{u} + i\vec{v} \rangle)) = \\ & \stackrel{**}{=} \frac{1}{4} (\cancel{\langle \vec{u}, \vec{u} \rangle} - \langle \vec{u}, i\vec{v} \rangle - \langle i\vec{v}, \vec{u} \rangle + \cancel{\langle i\vec{v}, i\vec{v} \rangle} - (\cancel{\langle \vec{u}, \vec{u} \rangle} + \langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle + \cancel{\langle i\vec{v}, i\vec{v} \rangle})) = \\ & = \frac{1}{4} (-2(\langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle)) \stackrel{***}{=} -\frac{1}{2} (i\langle \vec{u}, \vec{v} \rangle - i\overline{\langle \vec{u}, \vec{v} \rangle}) \stackrel{****}{=} -\frac{1}{2} i (i 2 \operatorname{Im} \langle \vec{u}, \vec{v} \rangle) = \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) \\ & \quad -\frac{1}{2} i \cdot 2i = -\frac{2}{2} i^2 = -1 \cdot (-1) = 1 \end{aligned}$$

* - skalarovita k 1. slovice tj. $\forall \vec{u}, \vec{v}, \vec{w} \in V: \langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$
 $t \in \mathbb{C} \quad \langle t \cdot \vec{u}, \vec{v} \rangle = \overline{t} \langle \vec{u}, \vec{v} \rangle$

** - linearity k 2. slovice tj. $\forall \vec{u}, \vec{v}, \vec{w} \in V: \langle \vec{u}, \vec{v} + \vec{w} \rangle = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle$
 $t \in \mathbb{C} \quad \langle \vec{u}, t \cdot \vec{v} \rangle = t \langle \vec{u}, \vec{v} \rangle$

*** - vlastnost komplexních čísel $x + \bar{x} = 2 \operatorname{Re} x$
 $y - \bar{y} = 2i \operatorname{Im} y$